

A Professional Forecasting Programme for an Electricity-Market Company

From raw data to monitored production forecasts

Project: Alqueva PSP–PV–BESS 24-hour energy trading optimizer

Document status: plan for review | Date: 10 October 2026

1 Purpose and how to read this plan

This document describes, step by step, how a professional energy-market team would build, validate, deploy and maintain machine-learning forecasts, and how we will apply the same process to the forecasters of this project. It is deliberately complete: no step is shortened, because the work can be done on a computer without a deadline.

The plan has six parts (A to F) and eighteen phases (0 to 17). Every phase lists its steps and an *exit criterion*: the evidence that must exist before the next phase starts. We complete the phases in order, one at a time.

What the literature and industry practice say

- **Benchmark honestly.** The standard review of day-ahead price forecasting (Lago et al., 2021) identifies the weak points of the field: short or single-market test samples, no comparison with simple benchmarks, unsuitable accuracy metrics and no test of statistical significance.
- **Backtest the way production will run.** Validation always trains on the past and tests on the future with rolling origins; shuffled cross-validation leaks information. A model that cannot beat a naive benchmark has not found structure in the data.
- **Probabilistic forecasts need proper scoring.** Quantile forecasts are scored with the pinball loss; the CRPS is approximated by averaging pinball losses over a quantile grid; coverage is tested separately; models are compared with Diebold–Mariano tests. Decision-based metrics (profit) are added because a better CRPS does not always mean better decisions.
- **Balancing markets are harder.** Studies of German aFRR and imbalance prices find no universally best model and poor results for activated volumes; Nordic mFRR studies find that extra model or feature complexity does not reliably help. We found no Iberian study, so our own evidence is needed.
- **Production needs MLOps.** Versioned features, a model registry, data validation, drift monitoring and champion–challenger promotion are standard in energy-trading forecasting systems.

Where the project stands today

The following results were measured earlier on real market data (last twelve months as the test window).

Forecast	Result versus the naive benchmark
Day-ahead price	about 2.6% better
IDA1–3 and XBID	1% to 5% worse
aFRR capacity price	very poor (no one-day lag feature)
mFRR price	about 0.2% better
PV (irradiance and temperature)	worse than naive
Reservoir inflow	worse than naive

Existing foundations: real data rebuilt from OMIE and REN, leakage fixes, walk-forward cross-validation, naive baselines, autoregressive evaluation of the intraday forecasters. Missing: probabilistic forecasts, fundamental features (load, wind, solar, fuel prices), significance tests, experiment tracking, drift monitoring, a forecast archive and decision-value evaluation.

2 Part A — Foundations

2.1 Phase 0. Charter and decision mapping

- List every decision that consumes a forecast: day-ahead bid, IDA1–3 and XBID trades, aFRR and mFRR capacity offers, PV and inflow planning, risk limits.
- For each forecast define the **as-of time**: exactly which information exists at the moment of the decision (day-ahead gate 12:00 CET, IDA1 15:00 CET, IDA2 22:00 CET, IDA3 10:00 CET, reserve gates, and so on). This is the main protection against leakage.
- Define horizon, resolution (hourly and 15-minute), output format (point, quantiles or scenarios) and the consumers of each forecast.
- Define success in euros (value of the forecast in the optimiser), not only in MAE.
- Fix the forecast catalogue: targets, owners, update schedule, fallback.

Exit criterion. A signed-off forecast catalogue with an as-of time and a consumer for every target.

2.2 Phase 1. Data inventory and access

- For every source (OMIE prices, REN reserve data, weather observations and forecasts, TSO fundamentals, fuel and carbon prices, reservoir data): history length, update delay, licence, API limits, cost.
- Record **vintages**: the forecast that was issued at the time, not only the final actual value (point-in-time data), for load, wind, solar and weather forecasts.
- Build a data dictionary (variable, unit, resolution, time zone, source, definition) and a lineage map.
- Identify gaps and substitutes, and the legal use of each dataset.

Exit criterion. Every input has an owner, a timestamp rule, a licence note and a quality note.

2.3 Phase 2. Ingestion and storage

- Three zones: *raw* (immutable, as downloaded), *cleaned*, and *feature-ready*, each versioned.
- Incremental, repeatable loads: retries, idempotent writes, checksums, load logs.
- Time handled properly: UTC internally; 23- and 25-hour days; 92- and 100-period quarter-hour

days; the move from hourly to 15-minute products (2025); the 2024 intraday-auction reform; the 2026 aFRR platform change. Each is stored as a **regime marker**.

- Data contracts: schema, types, ranges and freshness rules checked on every load.

Exit criterion. Any past date can be rebuilt exactly from the raw zone.

2.4 Phase 3. Data quality and cleaning

- Automated checks: schema, ranges, duplicates, gaps, stale feeds, unit consistency, DST artefacts.
- Price spikes and negative prices are information: flag them, never delete them blindly.
- Written imputation rules per variable that use only information available at the as-of time.
- Keep the provenance label (live, synthetic) on every row; synthetic rows are excluded from training and testing.
- A daily data-quality score and an automatic report.

Exit criterion. A quality report and a pass/fail gate that must be green before any model run.

2.5 Phase 4. Exploratory analysis and market understanding

- Seasonality (daily, weekly, annual), autocorrelation, stationarity, structural breaks (the 2022 energy crisis, solar price cannibalisation, the April 2025 blackout effects).
- Relationships between markets: day-ahead, IDA1–3, XBID, reserves, imbalance.
- Frequency of spikes and negative prices, distribution shape, heteroscedasticity.
- Effective sample size after the regime changes; which history is still representative.

Exit criterion. A written findings note that justifies the modelling choices of the next parts.

3 Part B — Measurement before modelling

3.1 Phase 5. Problem formulation per target

- Output type: point, quantile or scenario; direct versus recursive multi-step; per-hour models versus a full 24-hour vector.
- Targets as price levels or as spreads to the day-ahead price (important for intraday).
- A variance-stabilising transform that allows negative prices (for example the inverse hyperbolic sine).
- Consistency between hourly and 15-minute forecasts.
- The information set per target, written down and tested.

3.2 Phase 6. Evaluation framework (the contract, built first)

- Rolling-origin backtests with a realistic retraining schedule and a time gap between training and test.
- A **locked final test period** that nobody tunes on.
- Point metrics: MAE, RMSE and scaled errors against the naive forecast (percentage errors are avoided because prices can be near zero or negative).
- Probabilistic metrics: pinball loss on a stated quantile grid, CRPS, coverage tests, interval scores, PIT histograms.
- Significance: Diebold–Mariano tests with a HAC variance, corrected for testing many hours

and models.

- Benchmarks per target: same hour yesterday, same hour last week, a simple regression (ARX), then the candidates.
- **Decision value:** profit of the Alqueva optimiser when fed each forecast (naive, model, perfect foresight).
- Automated leakage tests and reproducibility tests.

Exit criterion. One command scores any forecaster in the same way and produces the same report.

4 Part C — Modelling

4.1 Phase 7. Feature engineering and feature store

- Calendar and holidays for Portugal and Spain, bridge days, DST.
- Lags and rolling statistics that respect the as-of time.
- Fundamentals: load, wind and solar forecasts, residual load, gas and carbon prices, interconnection, hydro levels, weather forecasts.
- A feature catalogue with each feature’s availability time, and point-in-time correctness tests.
- Feature selection with stability checks across periods; versioned feature definitions.

4.2 Phase 8. Baselines and candidate models

- Statistical: seasonal naive, ARX, LASSO-estimated ARX with several calibration windows (a strong benchmark in the review), generalised additive models.
- Machine learning: gradient boosting (LightGBM, XGBoost), random forest, quantile boosting.
- Deep learning: feed-forward networks, LSTM and GRU, N-BEATS, Temporal Fusion Transformer, distributional networks.
- Foundation models used zero-shot, as a benchmark only.
- Rule: a complex model must beat the simple one significantly, otherwise it is dropped.

4.3 Phase 9. Training, tuning and ensembling

- Hyper-parameters tuned inside the rolling backtest (nested, time-aware), never on the test period.
- Forecast combinations: simple averages often win and are tested first.
- Comparison of calibration-window lengths and retraining frequencies.

4.4 Phase 10. Probabilistic forecasts and scenarios

- Quantile models, quantile regression averaging, distributional networks.
- Conformal prediction for reliable intervals (recent studies apply it to day-ahead and balancing prices).
- Scenario generation that keeps the correlation across hours, feeding the stochastic CVaR bidding model.
- Calibration diagnostics and automatic recalibration.

Exit criterion. Intervals reach their nominal coverage out of sample, and scenarios pass the correlation checks.

5 Part D — Target-specific work

5.1 Phase 11. One workstream per target

Target	Focus
Day-ahead price	Biggest revenue lever; the full pipeline is built here first (Phases 5–10).
IDA1–3 and XBID	Model the spread to the day-ahead price; use the outcome of the earlier gate as information; market-microstructure features.
aFRR and mFRR capacity prices	Day-before lag, merit-order features, bid and award behaviour; handle the 2026 product and platform change as a regime; expect modest gains.
Imbalance and activation	Probabilistic, low expectations; value is mainly in risk estimates.
PV	Forecast irradiance and temperature with weather-model post-processing, then apply the physics model; compare with satellite irradiance; probabilistic.
Reservoir inflow	Hydrological view (rainfall, runoff, irrigation rules); find out why the current forecast is worse than naive.

6 Part E — Industrialisation

6.1 Phase 12. Experiment tracking and reproducibility

- Every run records code version, data snapshot, configuration, random seed, metrics and plots.
- Automatic model cards (purpose, data, metrics, limits).

6.2 Phase 13. Model selection, governance and acceptance gates

- A model is promoted only if it beats the benchmark with statistical significance *and* adds decision value.
- Champion–challenger rule, documented assumptions and a written fallback.
- Review and sign-off record for each promotion.

6.3 Phase 14. Production pipeline

- Scheduling tied to each market gate, retries, data-freshness limits, automatic fallback to a simple forecast when data are late.
- A **forecast archive** with timestamps (essential for later evaluation).
- Tests in continuous integration, containers, a fixed validated interface to the optimiser.

6.4 Phase 15. Monitoring and maintenance

- Input drift (population stability index, Kolmogorov–Smirnov), error drift, bias and calibration drift computed on realised data.
- Alert thresholds, a retraining policy (scheduled and triggered), shadow runs of challengers.
- Incident runbook and a periodic review.

7 Part F — Value and communication

7.1 Phase 16. Decision value

- Profit with perfect foresight versus model versus naive forecasts in the optimiser.
- Sensitivity of profit to forecast error and the value in euros of each improvement.
- Stochastic bidding with quantile forecasts versus point forecasts.

7.2 Phase 17. Documentation and reporting

- Methodology note, model cards, and the existing dashboard page extended with these metrics.
- A short narrative of the work for technical interviews.

8 Proposed order of work

1. Phases 0 and 1, plus an **audit** of today’s forecasters (no new modelling code).
2. Phases 2 to 4: data layer, data quality, exploratory analysis.
3. Phase 6: the evaluation framework, before any new model.
4. The day-ahead price end to end (Phases 5 and 7 to 10), then reserves, then intraday, PV and inflow.
5. Phases 12 to 15, then Phase 16 to show the value.

9 Decisions needed before starting

1. **Order of targets.** Proposed: day-ahead price, then aFRR and mFRR, then intraday, PV and inflow.
2. **Data access.** Are Spanish and Portuguese TSO data (load, wind and solar forecasts) and a weather forecast source available, or only freely available data?
3. **Tools.** Open-source Python (pandas, scikit-learn, LightGBM, Optuna, MLflow-style tracking) or specific tools?
4. **Tracking.** This document is the master checklist; each completed phase is recorded with its exit evidence.

References

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